Update Joint Modeling 12/10

Overview methods

Table 1: Configuration of the methods; N is the number of clusters in the fit.

Method	Configuration
Classical ML	`lme4::lmer`, ML (REML = FALSE); Wald t test on β_3 with $N-2$ df
MI + closed form	`mice` `2l.pmm`, 3 imputations, `maxit` = 10; closed form (CbC) per imputation, pooled with Rubin's rules; Wald t test on β_3 with Barnard-Rubin df (complete-data df $N-2$)
Reweighting	Closed form, reweighted until max $\Delta\beta < 10^{-6}$, at most 30 iterations; Wald t test on β_3 with $N-2$ df
LSPIM	Pseudo-score GEE model (Firth-penalised, Ford-Westgate small-sample sandwich), fitted with an improved algorithm; Holm-adjusted test that the per-visit treatment effects differ; run at every n

Table 2: Definition of "not converged" per method.

Method	"Not converged" when
Classical ML	optimizer code $\neq 0$ or an lme4 convergence message (e.g. max|grad| > 0.002)
MI + closed form	never; a failed fit in any imputation counts as an error
Reweighting	the loop reaches 30 iterations without meeting the $\Delta\beta$ criterion
LSPIM	the penalised GEE does not converge (fitted probabilities numerically 0 or 1)

Each replicate gets exactly one of five convergence statuses, checked in this order:

Table 3: Convergence statuses.

Status	Meaning
Error	The fit failed (no estimates); excluded from all metrics.
Not converged	Fit returned estimates, but the method's convergence criterion (table above) was not met.
Converged (singular)	Converged, but the random-effects covariance is at the boundary (smallest eigenvalue $\leq 10^{-6}$, or the final D repaired to positive definite).
Converged (warning)	Converged and not singular, but a warning was raised (e.g. lme4's boundary message at its looser tolerance; for reweighting NaNs in an intermediate iteration; for LSPIM the combined covariance V repaired to the nearest positive semi-definite matrix).
Converged	Converged, not singular, no warning.

- All tests at $\alpha = 0.05$; coverage of the 95% t interval for β_3.
- All statuses except "error" are kept in all metrics.
- LSPIM gives only a test decision, no estimate of β_3: it appears in the type I error, power, convergence and computation time results, not in MSE, coverage or fraction of missing information. Its errors and warnings are described in the appendix.

Simulation setup

Linear model, $t_j = j = 0, \ldots, 11$:

$$y_{ij} = \beta_0 + \beta_1 Treatment_i + \beta_2 t_j + \beta_3 Treatment_i \times t_j + b_{0i} + b_{1i} t_j + \varepsilon_{ij}$$

Log model, $t_j = j = 0, \ldots, 11$:

$$y_{ij} = \beta_0 + \beta_1 Treatment_i + \beta_2 \log(1 + t_j) + \beta_3 Treatment_i \times \log(1 + t_j) + b_{0i} + b_{1i} \log(1 + t_j) + \varepsilon_{ij}$$

In both models $(b_{0i}, b_{1i})^\top \sim N(0, D)$ with $D = \begin{pmatrix} d_{11} & d_{12} \\ d_{12} & d_{22} \end{pmatrix}$ and $\varepsilon_{ij} \sim N(0, \sigma^2)$.

Table 4: Data-generating parameters; a / b / c gives the values of the large, small and null effect.

Effect	Symbol	Linear	Log
Intercept	β_0	2.4562	2.4562
Treatment effect	β_1	0	-2.42 / -0.385 / 0
Time effect	β_2	0.2792	1.2359
Treatment × time	β_3	0.22 / 0.035 / 0	1.948 / 0.310 / 0
Random intercept variance	d_{11}	7.3174	7.3174
Random slope variance	d_{22}	0.2239	4.3875
Random effects covariance	d_{12}	-0.4985	-2.2067
Residual variance	σ^2	3.1508	3.1508

- Linear parameters taken from the toenail data ($\beta_3 = 0.035$); $\beta_3 = 0.22$ added as a larger effect.
- Log parameters: β_2, d_{22} and d_{12} rescaled by $11/\log 12$, so the rise and the slope variance at $t = 11$ match the linear scenario.
- Log crossing: arms cross at $t \approx 2.5$; the treatment difference at $t = 11$ equals that of the linear scenario with $\beta_3 = 0.22$ (large) or 0.035 (small).
- Log null: identical arms ($\beta_1 = \beta_3 = 0$).

Table 5: Simulation design.

Variable	Value
Number of clusters	10, 20, 50, 100, 1000
Number of repeated measures	12
Missingness pattern	"Half-missing" (HM): half of the clusters have no missingness, the other half have a uniform probability to keep 1–11 observations; dropout is monotone. "Three observations minimum" (3O): as half-missing, but with at least three observations per cluster.
Simulations per scenario	5000

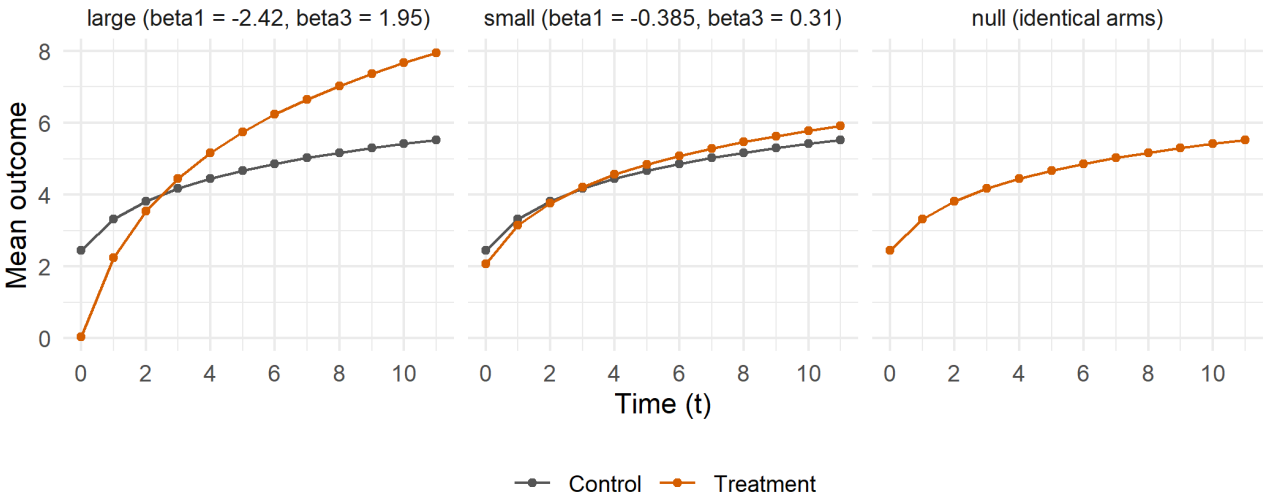


Figure 1: Mean profiles per arm for the log time trend, from the simulation parameters.

Results - linear time trend

Mean squared error of β_3

Table 6: MSE of beta3 (HM = half-missing, 3O = three observations minimum; columns = true beta3).

n	Method	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	Classical ML	0.137	0.138	0.135	0.125	0.125	0.126
10	MI + closed form	0.122	0.119	0.117	0.113	0.110	0.111
10	Reweighting	0.136	0.138	0.134	0.124	0.125	0.125
20	Classical ML	0.0640	0.0636	0.0673	0.0595	0.0631	0.0627
20	MI + closed form	0.0627	0.0621	0.0651	0.0588	0.0606	0.0608
20	Reweighting	0.0650	0.0643	0.0684	0.0594	0.0630	0.0626
50	Classical ML	0.0245	0.0253	0.0248	0.0241	0.0232	0.0236
50	MI + closed form	0.0255	0.0266	0.0258	0.0254	0.0240	0.0245
50	Reweighting	0.0249	0.0256	0.0252	0.0241	0.0232	0.0236
100	Classical ML	0.0123	0.0128	0.0122	0.0119	0.0117	0.0115
100	MI + closed form	0.0130	0.0134	0.0131	0.0126	0.0123	0.0121
100	Reweighting	0.0125	0.0129	0.0124	0.0119	0.0118	0.0115
1000	Classical ML	0.00129	0.00125	0.00125	0.00117	0.00119	0.00121
1000	MI + closed form	0.00141	0.00135	0.00136	0.00126	0.00131	0.00131
1000	Reweighting	0.00132	0.00127	0.00127	0.00117	0.00119	0.00121

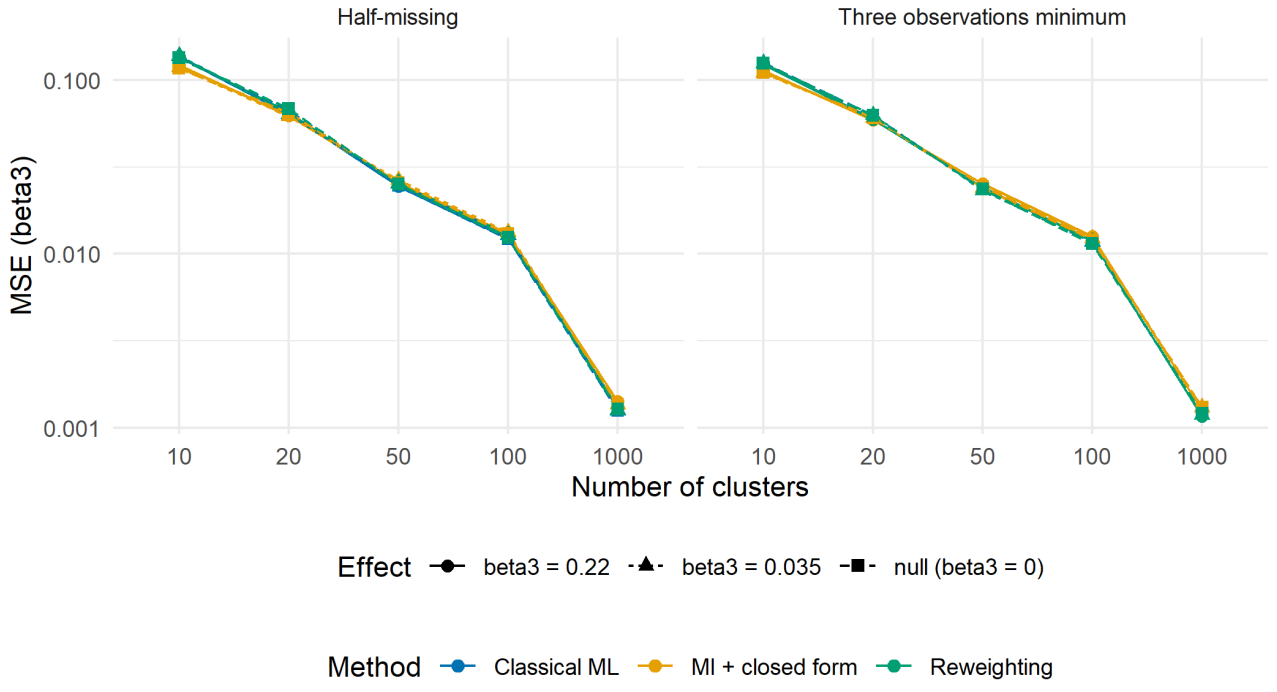


Figure 2: MSE of beta3 for the linear time trend (log scale).

- MSE falls with the number of clusters at the same rate for all methods; the effect size hardly matters, and MSE is slightly lower under three observations minimum.
- MI has the lowest MSE at $n = 10$ and 20; from $n = 50$ onwards classical ML and reweighting are slightly better and nearly identical.

Type I error

Table 7: Type I error rate of the interaction test (beta3 = 0).

n	Method	HM: 0	3O: 0
10	Classical ML	0.102	0.0988
10	MI + closed form	0.0466	0.0392
10	Reweighting	0.0596	0.0648
10	LSPIM	0.109	0.123
20	Classical ML	0.0780	0.0722
20	MI + closed form	0.0518	0.0490
20	Reweighting	0.0610	0.0606
20	LSPIM	0.0775	0.0813
50	Classical ML	0.0562	0.0546
50	MI + closed form	0.0404	0.0410
50	Reweighting	0.0500	0.0512
50	LSPIM	0.0372	0.0374
100	Classical ML	0.0532	0.0528
100	MI + closed form	0.0430	0.0400
100	Reweighting	0.0500	0.0496
100	LSPIM	0.0370	0.0352
1000	Classical ML	0.0472	0.0494
1000	MI + closed form	0.0472	0.0460
1000	Reweighting	0.0478	0.0488
1000	LSPIM	0.0398	0.0446

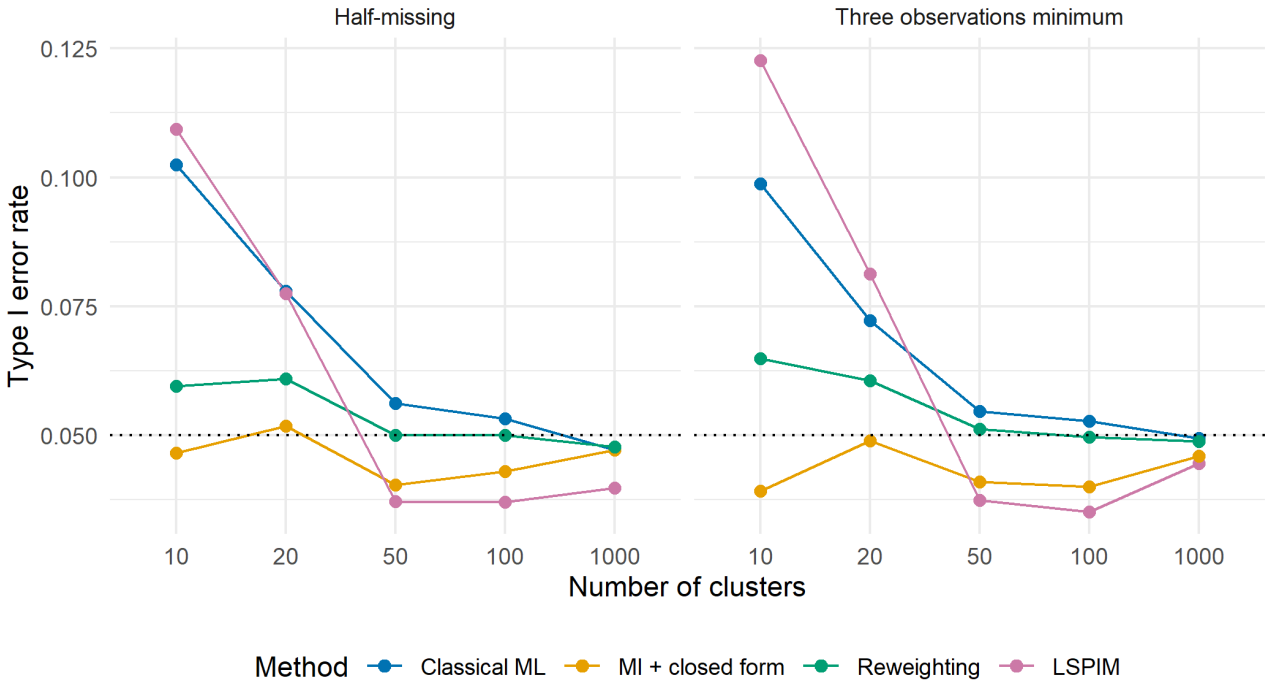


Figure 3: Type I error rate for the linear time trend (dotted line: nominal 0.05).

- Classical ML is still anti-conservative at small n (about 0.10 at $n = 10$, 0.07–0.08 at $n = 20$) and close to nominal from $n = 50$.
- Reweighting is slightly inflated at $n = 10$ and 20 (about 0.06) and nominal from $n = 50$.

- MI is at or slightly below nominal at every n (about 0.04–0.05).
- LSPIM is the most inflated at $n = 10$ (0.11–0.12) and still at $n = 20$ (about 0.08); from $n = 50$ onwards it is conservative (0.035–0.045).

Power

Table 8: Power of the interaction test.

n	Method	HM: 0.22	HM: 0.035	3O: 0.22	3O: 0.035
10	Classical ML	0.155	0.105	0.140	0.100
10	MI + closed form	0.0708	0.0436	0.0596	0.0372
10	Reweighting	0.0930	0.0626	0.0963	0.0635
10	LSPIM	0.120	0.117	0.123	0.126
20	Classical ML	0.165	0.0702	0.165	0.0756
20	MI + closed form	0.112	0.0472	0.108	0.0446
20	Reweighting	0.141	0.0574	0.140	0.0650
20	LSPIM	0.130	0.0768	0.119	0.0773
50	Classical ML	0.290	0.0648	0.295	0.0568
50	MI + closed form	0.220	0.0546	0.215	0.0422
50	Reweighting	0.267	0.0586	0.282	0.0528
50	LSPIM	0.103	0.0394	0.104	0.0414
100	Classical ML	0.522	0.0654	0.528	0.0642
100	MI + closed form	0.397	0.0582	0.416	0.0486
100	Reweighting	0.500	0.0616	0.518	0.0616
100	LSPIM	0.181	0.0434	0.197	0.0462
1000	Classical ML	1.00	0.162	1.00	0.174
1000	MI + closed form	0.998	0.130	1.00	0.142
1000	Reweighting	1.00	0.158	1.00	0.173
1000	LSPIM	0.990	0.0726	0.991	0.0742

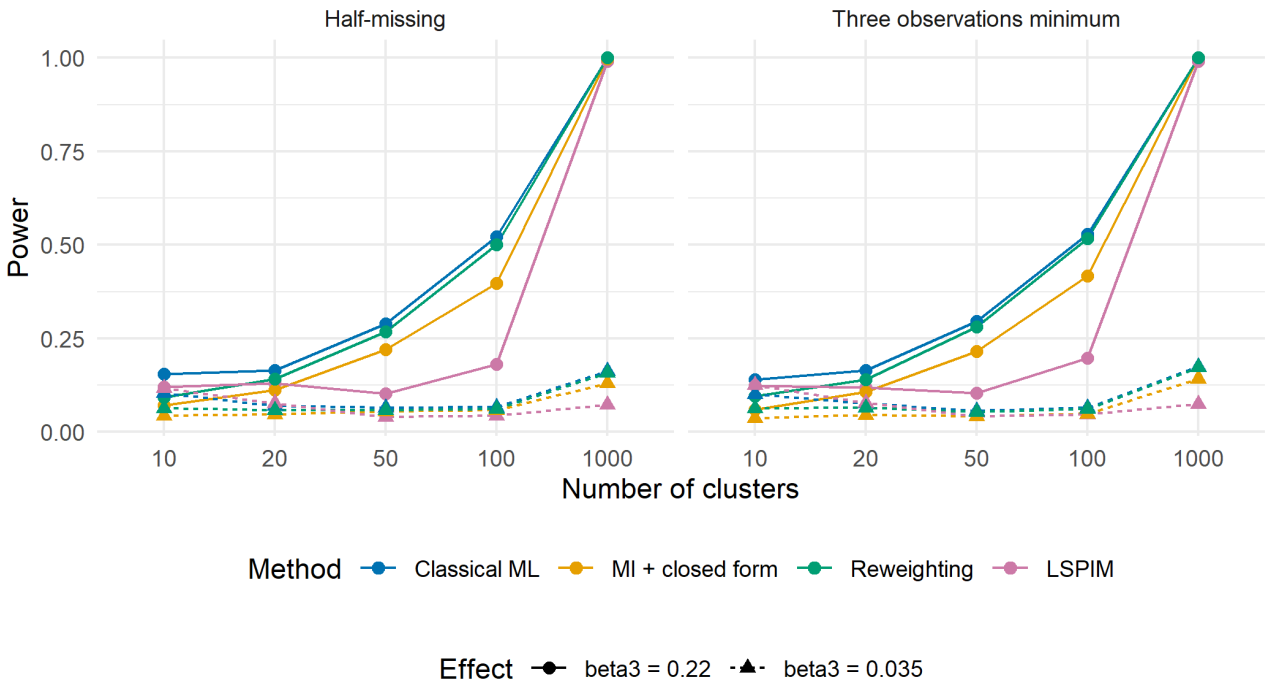


Figure 4: Power for the linear time trend.

- Classical ML has the highest power, then reweighting, then MI; at small n the lead of classical ML partly reflects its type I error inflation.

- For $\beta_3 = 0.22$ classical ML, reweighting and MI reach full power at $n = 1000$; for $\beta_3 = 0.035$ power stays low even at $n = 1000$ (0.13–0.17), lowest for MI.
- LSPIM has much lower power than the other methods for a linear effect: for $\beta_3 = 0.22$ about 0.18–0.20 at $n = 100$ (others 0.40–0.53), 0.99 at $n = 1000$; for $\beta_3 = 0.035$ it stays near its type I error level. At $n = 10$ and 20 its power mostly reflects its type I error inflation.

Coverage of β_3

Table 9: Coverage of the 95% t interval for beta3.

n	Method	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	Classical ML	0.896	0.895	0.898	0.907	0.904	0.901
10	MI + closed form	0.956	0.955	0.953	0.961	0.963	0.961
10	Reweighting	0.940	0.939	0.940	0.942	0.937	0.935
20	Classical ML	0.926	0.930	0.922	0.935	0.924	0.928
20	MI + closed form	0.952	0.954	0.948	0.959	0.959	0.951
20	Reweighting	0.939	0.947	0.939	0.945	0.937	0.939
50	Classical ML	0.948	0.939	0.944	0.943	0.949	0.945
50	MI + closed form	0.956	0.951	0.960	0.953	0.960	0.959
50	Reweighting	0.950	0.946	0.950	0.944	0.952	0.949
100	Classical ML	0.948	0.946	0.947	0.945	0.945	0.947
100	MI + closed form	0.958	0.950	0.957	0.958	0.957	0.960
100	Reweighting	0.952	0.949	0.950	0.946	0.947	0.950
1000	Classical ML	0.950	0.952	0.953	0.956	0.947	0.951
1000	MI + closed form	0.952	0.960	0.953	0.962	0.957	0.954
1000	Reweighting	0.950	0.953	0.952	0.956	0.947	0.951

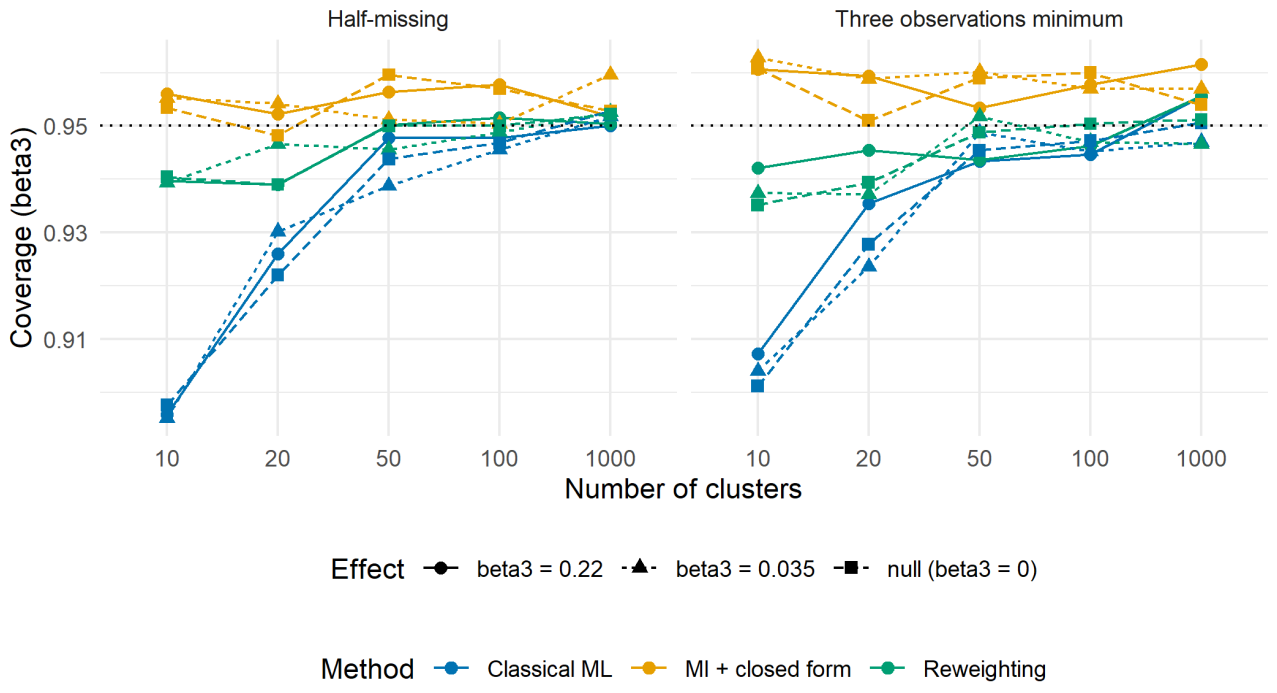


Figure 5: Coverage of the 95% interval for beta3 for the linear time trend (dotted line: 0.95).

- Classical ML undercovers at small n (about 0.90 at $n = 10$, 0.92–0.93 at $n = 20$) and is close to nominal from $n = 50$.
- Reweighting is slightly below nominal at $n = 10$ and 20 (0.94) and nominal from $n = 50$.
- MI is at or slightly above nominal at every n (0.95–0.96).

Convergence

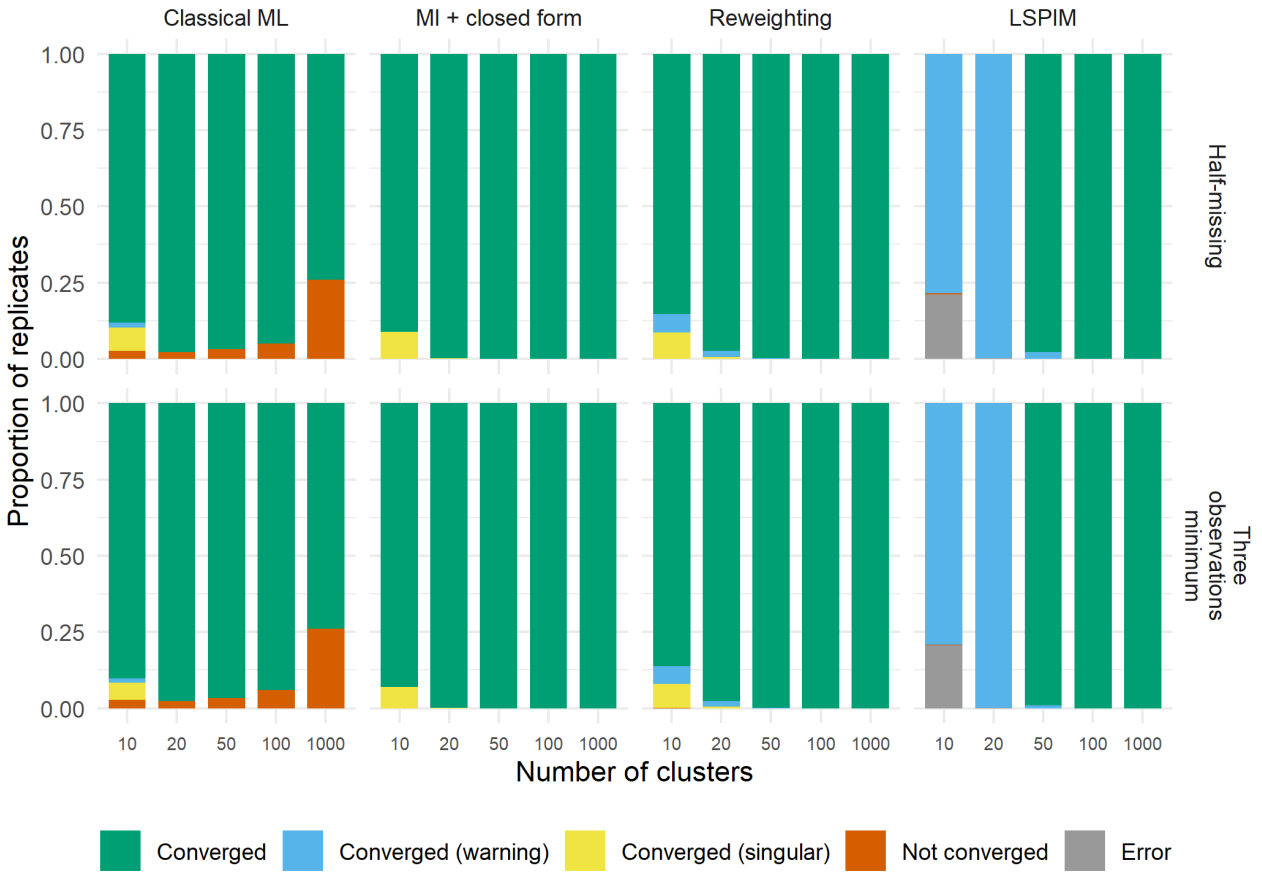


Figure 6: Convergence status for the linear time trend, pooled over effect sizes.

- Singular fits occur at $n = 10$ for classical ML, MI and reweighting at a similar rate (5–9%) and almost vanish from $n = 20$.
- Reweighting and MI essentially always converge; reweighting has a small warning share at $n = 10$ and 20.
- Classical ML has a small "not converged" share at every n, which grows to about 26% at $n = 1000$. This is a false positive of lme4's absolute gradient check, which scales with the sample size: the flagged fits are kept in all metrics.
- LSPIM: about 20% errors at $n = 10$, and almost all fits at $n = 10$ and 20 carry a warning; see the appendix.

Computation time

Table 10: Median computation time per replicate (seconds).

n	Method	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	Classical ML	0.0180	0.0170	0.0180	0.0160	0.0160	0.0170
10	MI + closed form	1.16	1.04	1.04	1.05	1.05	1.04
10	Reweighting	0.0560	0.0390	0.0430	0.0490	0.0470	0.0500
10	LSPIM	0.0150	0.00800	0.00800	0.00900	0.00900	0.00900
20	Classical ML	0.0220	0.0150	0.0180	0.0200	0.0190	0.0200
20	MI + closed form	1.26	1.14	1.14	1.15	1.15	1.15
20	Reweighting	0.111	0.0890	0.0900	0.102	0.103	0.102
20	LSPIM	0.0420	0.0130	0.0130	0.0140	0.0140	0.0140
50	Classical ML	0.0260	0.0180	0.0210	0.0225	0.0180	0.0190
50	MI + closed form	1.58	1.44	1.43	1.45	1.45	1.45
50	Reweighting	0.288	0.231	0.234	0.267	0.264	0.266
50	LSPIM	0.234	0.0490	0.0490	0.0590	0.0570	0.0550
100	Classical ML	0.0340	0.0300	0.0290	0.0290	0.0300	0.0290
100	MI + closed form	2.17	1.97	1.96	1.99	2.00	2.00
100	Reweighting	0.609	0.511	0.509	0.570	0.569	0.572
100	LSPIM	1.26	0.218	0.214	0.263	0.253	0.251
1000	Classical ML	0.285	0.219	0.220	0.230	0.230	0.231
1000	MI + closed form	17.1	13.1	13.7	13.5	13.6	13.6
1000	Reweighting	15.8	14.7	14.7	16.4	16.4	16.4
1000	LSPIM	52.1	45.9	43.0	57.6	50.5	47.9

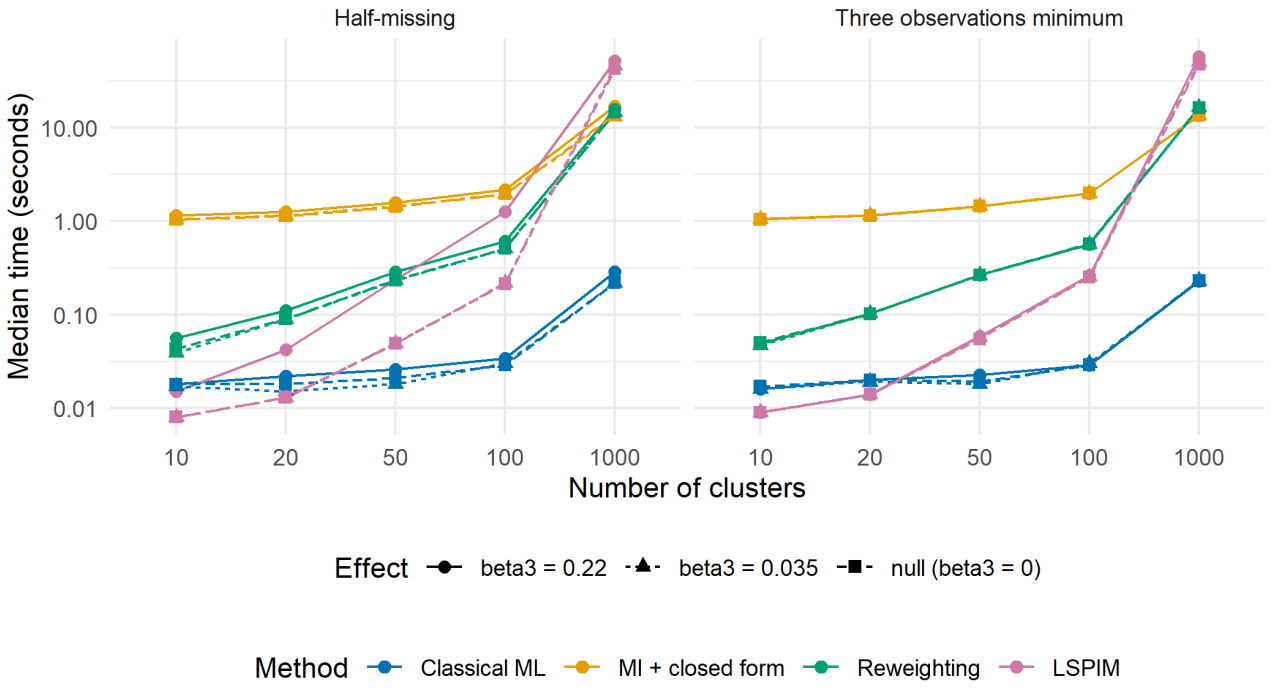


Figure 7: Median computation time per replicate for the linear time trend (log scale).

- Classical ML is fastest from $n = 50$ onwards; at $n = 10$ and 20 LSPIM is slightly faster.
- MI is slowest up to $n = 100$ (1–2 seconds per replicate).

- LSPIM is fast up to $n = 100$ (at most about 1 second) but grows roughly quadratically with n: at $n = 1000$ it is the slowest method (about 45–58 seconds), against about 14–17 seconds for MI and reweighting.

Fraction of missing information (MI)

Table 11: Mean fraction of missing information for beta3 (MI only).

n	HM: 0.22	HM: 0.035	HM: 0	3O: 0.22	3O: 0.035	3O: 0
10	0.231	0.223	0.226	0.212	0.211	0.205
20	0.252	0.245	0.246	0.230	0.231	0.223
50	0.264	0.262	0.268	0.242	0.239	0.239
100	0.275	0.271	0.271	0.248	0.244	0.243
1000	0.276	0.276	0.277	0.253	0.253	0.251

- The fraction of missing information is higher under half-missing than under three observations minimum, and increases slightly with n.

Results - log time trend

- MSE and coverage are not reported: the linear methods estimate a linear slope, which has no true value under the log trend.
- Effect sizes: large ($\beta_1 = -2.42$, $\beta_3 = 1.95$), small ($\beta_1 = -0.385$, $\beta_3 = 0.31$).

Type I error

Table 12: Type I error rate of the interaction test (identical arms).

n	Method	HM: 0	3O: 0
10	Classical ML	0.103	0.0976
10	MI + closed form	0.0392	0.0366
10	Reweighting	0.0496	0.0553
10	LSPIM	0.110	0.107
20	Classical ML	0.0762	0.0684
20	MI + closed form	0.0536	0.0454
20	Reweighting	0.0482	0.0430
20	LSPIM	0.0832	0.0731
50	Classical ML	0.0616	0.0560
50	MI + closed form	0.0508	0.0410
50	Reweighting	0.0436	0.0416
50	LSPIM	0.0390	0.0408
100	Classical ML	0.0560	0.0512
100	MI + closed form	0.0478	0.0440
100	Reweighting	0.0386	0.0354
100	LSPIM	0.0406	0.0406
1000	Classical ML	0.0498	0.0456
1000	MI + closed form	0.0432	0.0414
1000	Reweighting	0.0350	0.0338
1000	LSPIM	0.0422	0.0438

- Classical ML and LSPIM show the same small-n inflation as in the linear scenarios (about 0.10 and 0.11 at $n = 10$); LSPIM is again conservative from $n = 50$.
- MI is at or slightly below nominal at every n.
- Reweighting is close to nominal at $n = 10$ and 20, and becomes conservative from $n = 50$ onwards (about 0.035 at $n = 1000$).

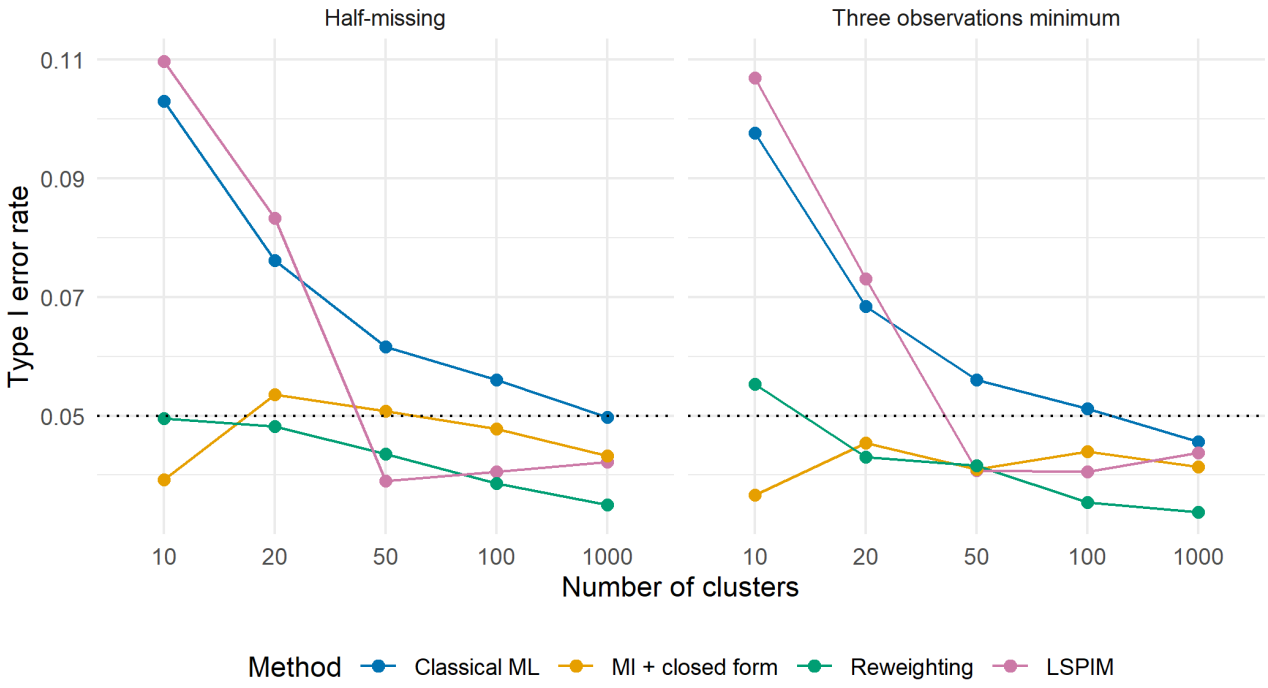
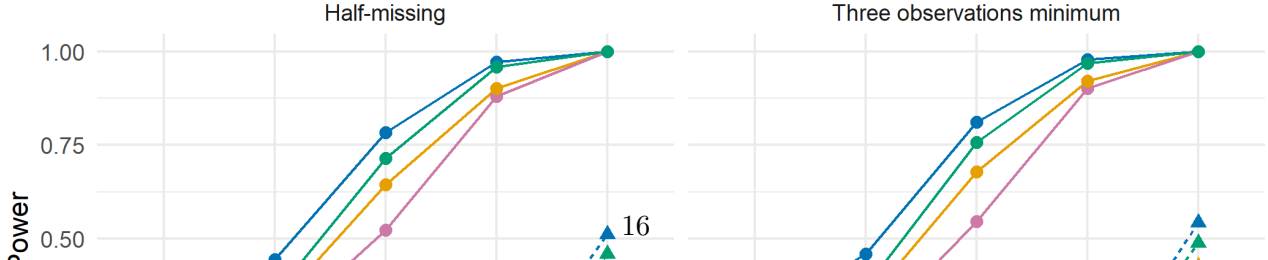


Figure 8: Type I error rate for the log time trend (dotted line: nominal 0.05).

Power

Table 13: Power of the interaction test.

n	Method	HM: large	HM: small	3O: large	3O: small
10	Classical ML	0.283	0.102	0.285	0.0902
10	MI + closed form	0.134	0.0414	0.141	0.0352
10	Reweighting	0.165	0.0482	0.183	0.0505
10	LSPIM	0.188	0.110	0.218	0.117
20	Classical ML	0.444	0.0806	0.458	0.0776
20	MI + closed form	0.315	0.0520	0.328	0.0472
20	Reweighting	0.349	0.0518	0.369	0.0556
20	LSPIM	0.254	0.0845	0.238	0.0776
50	Classical ML	0.783	0.0814	0.812	0.0832
50	MI + closed form	0.643	0.0638	0.678	0.0610
50	Reweighting	0.715	0.0612	0.757	0.0604
50	LSPIM	0.523	0.0490	0.546	0.0482
100	Classical ML	0.972	0.103	0.979	0.0946
100	MI + closed form	0.902	0.0798	0.921	0.0722
100	Reweighting	0.959	0.0762	0.969	0.0708
100	LSPIM	0.881	0.0576	0.901	0.0572
1000	Classical ML	1.00	0.511	1.00	0.543
1000	MI + closed form	1.00	0.409	1.00	0.426
1000	Reweighting	1.00	0.458	1.00	0.487
1000	LSPIM	1.00	0.288	1.00	0.293



- LSPIM is close to MI for the large effect (about 0.88–0.90 at $n = 100$), but has the lowest power for the small effect (about 0.29 at $n = 1000$, others 0.41–0.54).

Convergence

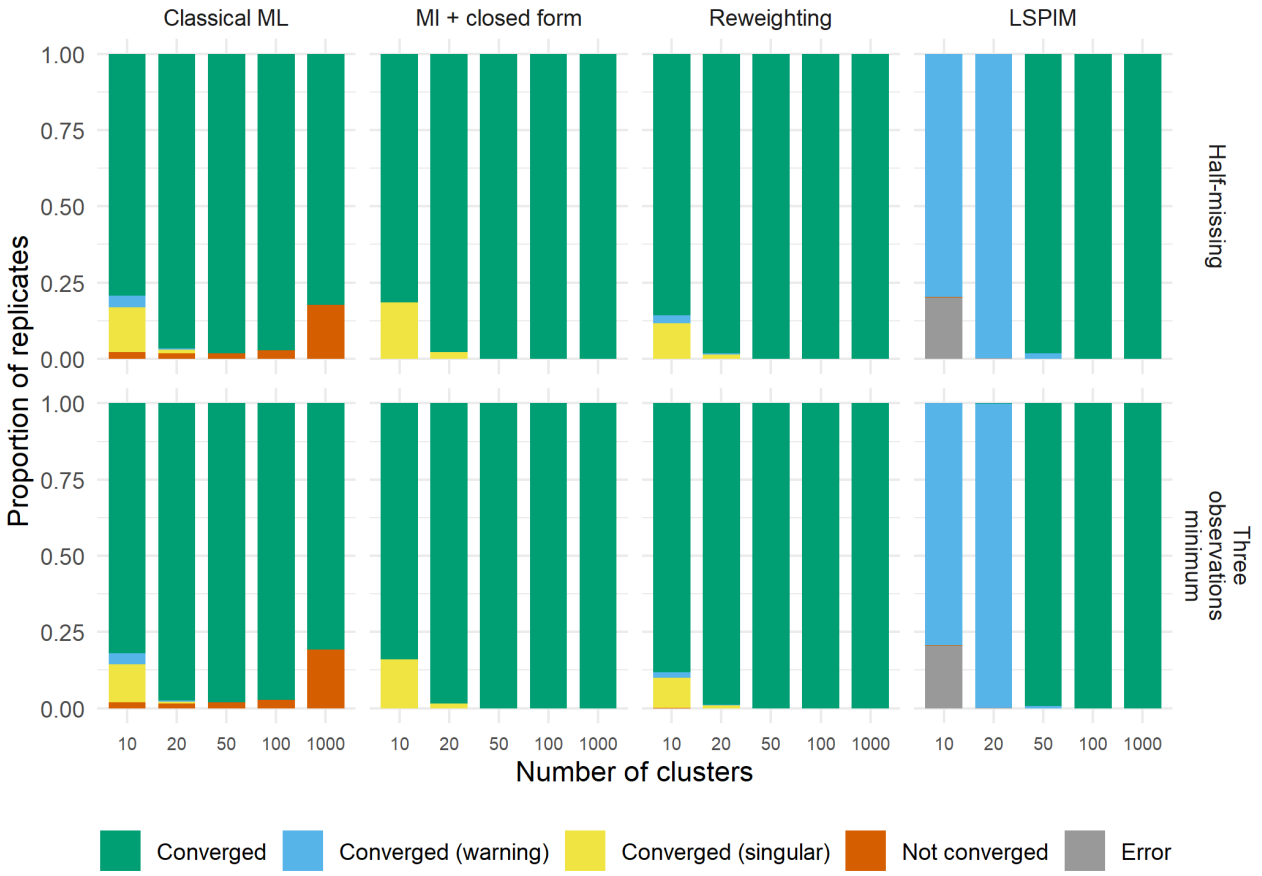


Figure 10: Convergence status for the log time trend, pooled over effect sizes.

- Same pattern as the linear scenarios, with more singular fits at $n = 10$ (10–20%).
- Classical ML's "not converged" share at $n = 1000$ (about 18%) is again the lme4 gradient false positive.
- LSPIM errors and warnings as in the linear scenarios; see the appendix.

Computation time

Table 14: Median computation time per replicate (seconds).

n	Method	HM: large	HM: small	HM: 0	3O: large	3O: small	3O: 0
10	Classical ML	0.0170	0.0170	0.0160	0.0140	0.0170	0.0160
10	MI + closed form	1.03	1.03	1.02	1.03	1.02	1.03
10	Reweighting	0.0410	0.0390	0.0430	0.0450	0.0470	0.0435
10	LSPIM	0.00800	0.00800	0.00800	0.00900	0.00900	0.00900
20	Classical ML	0.0170	0.0200	0.0190	0.0190	0.0190	0.0150
20	MI + closed form	1.13	1.13	1.12	1.13	1.13	1.13
20	Reweighting	0.0900	0.0900	0.0900	0.102	0.101	0.102
20	LSPIM	0.0130	0.0130	0.0130	0.0120	0.0140	0.0140
50	Classical ML	0.0210	0.0220	0.0170	0.0220	0.0220	0.0210
50	MI + closed form	1.41	1.41	1.40	1.41	1.42	1.41
50	Reweighting	0.232	0.230	0.231	0.265	0.263	0.265
50	LSPIM	0.0520	0.0480	0.0480	0.0600	0.0560	0.0560
100	Classical ML	0.0280	0.0270	0.0270	0.0290	0.0280	0.0280
100	MI + closed form	1.91	1.90	1.89	1.92	1.93	1.94
100	Reweighting	0.511	0.506	0.506	0.572	0.568	0.566
100	LSPIM	0.248	0.216	0.216	0.283	0.252	0.250
1000	Classical ML	0.197	0.207	0.202	0.212	0.219	0.219
1000	MI + closed form	13.1	13.5	12.7	13.2	13.2	13.3
1000	Reweighting	14.7	14.7	14.7	16.5	16.4	16.4
1000	LSPIM	57.3	44.3	41.2	64.2	50.3	46.7

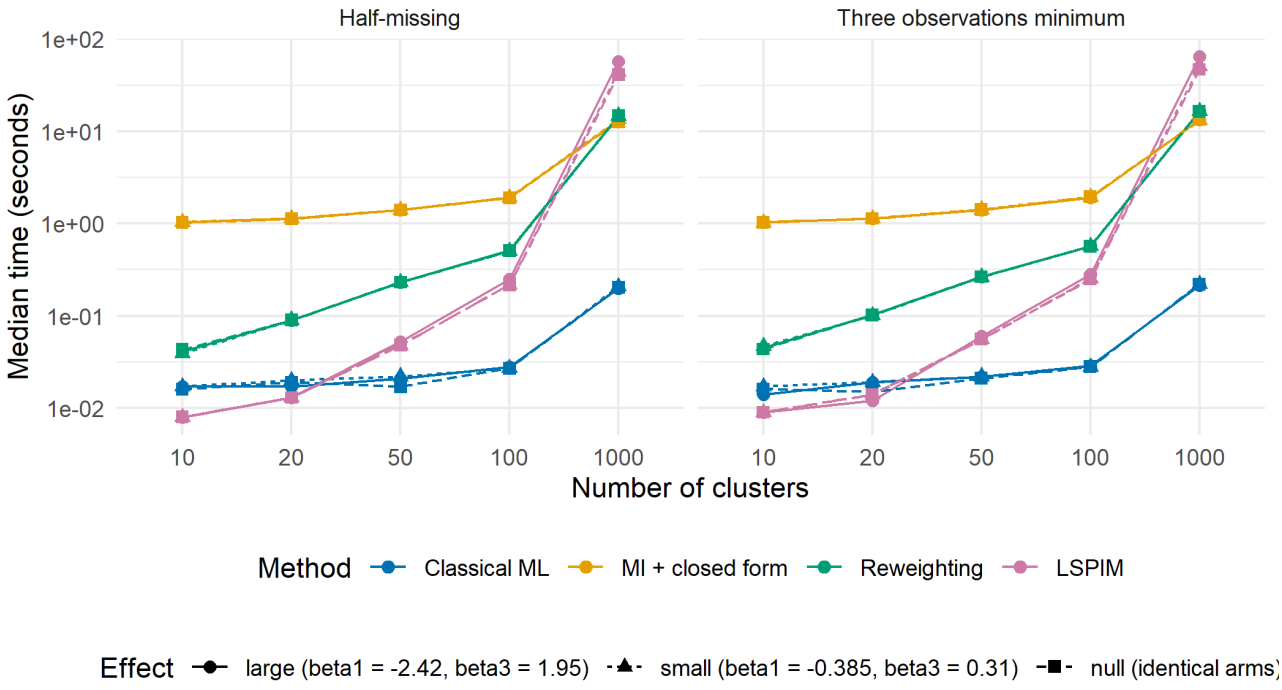


Figure 11: Median computation time per replicate for the log time trend (log scale).

- Same ordering as the linear scenarios.

Fraction of missing information (MI)

Table 15: Mean fraction of missing information for beta3 (MI only).

n	HM: large	HM: small	HM: 0	3O: large	3O: small	3O: 0
10	0.249	0.237	0.234	0.227	0.219	0.220
20	0.268	0.263	0.262	0.240	0.234	0.237
50	0.284	0.283	0.283	0.261	0.257	0.253
100	0.292	0.286	0.292	0.267	0.258	0.259
1000	0.302	0.296	0.303	0.270	0.270	0.270

- Slightly higher than in the linear scenarios; same pattern over dropout mechanism and n.

Suggested improvements

1. Classical ML convergence: replace lme4's absolute gradient flag with a relative gradient check, to remove the false "not converged" at large n.

Appendix: LSPIM errors and warnings

Table 16: LSPIM outcome per replicate (% of replicates, pooled over time trends and effect sizes). PSD = V repaired to the nearest PSD matrix; C1/C2 = leverage 1 in that clustering; zero col. = all-zero design column.

n	Dropout	Converged	Warning (PSD)	Not conv.	Error (C1)	Error (C2)	Error (zero col.)
10	HM	0.0	79.0	0.3	10.1	9.7	0.8
10	3O	0.0	79.3	0.2	9.8	9.9	0.8
20	HM	0.0	99.9	0.0	0.1	0.1	0.0
20	3O	0.0	99.9	0.0	0.1	0.0	0.0
50	HM	98.0	2.0	0.0	0.0	0.0	0.0
50	3O	99.1	0.9	0.0	0.0	0.0	0.0
100	HM	100.0	0.0	0.0	0.0	0.0	0.0
100	3O	100.0	0.0	0.0	0.0	0.0	0.0
1000	HM	100.0	0.0	0.0	0.0	0.0	0.0
1000	3O	100.0	0.0	0.0	0.0	0.0	0.0

- LSPIM uses three clusterings of the pairwise pseudo-observations: by left subject (C1), by right subject (C2) and by pair (C3); the combined covariance is $V = V_{C1} + V_{C2} - V_{C3}$.
- Errors occur almost only at $n = 10$ (about 20% of replicates):
  - about 10% each: a cluster has leverage 1 in clustering C1 or C2, so the Ford-Westgate small-sample correction is undefined;
  - about 1%: no between-arm pairs at one or more late visits (one arm has no observations left there), so the corresponding design columns are all zero.
- Warnings: at $n = 10$ and 20 nearly every fit that does not error has to repair the combined covariance V to the nearest positive semi-definite matrix (negative eigenvalues or non-positive variances); about 1–2% of fits at $n = 50$, and none from $n = 100$.
- Not converged: rare (below 1% at $n = 10$, almost none from $n = 20$); the penalised GEE gives fitted probabilities numerically 0 or 1.
- The repaired V at small n may contribute to LSPIM's type I error inflation at $n = 10$ and 20.